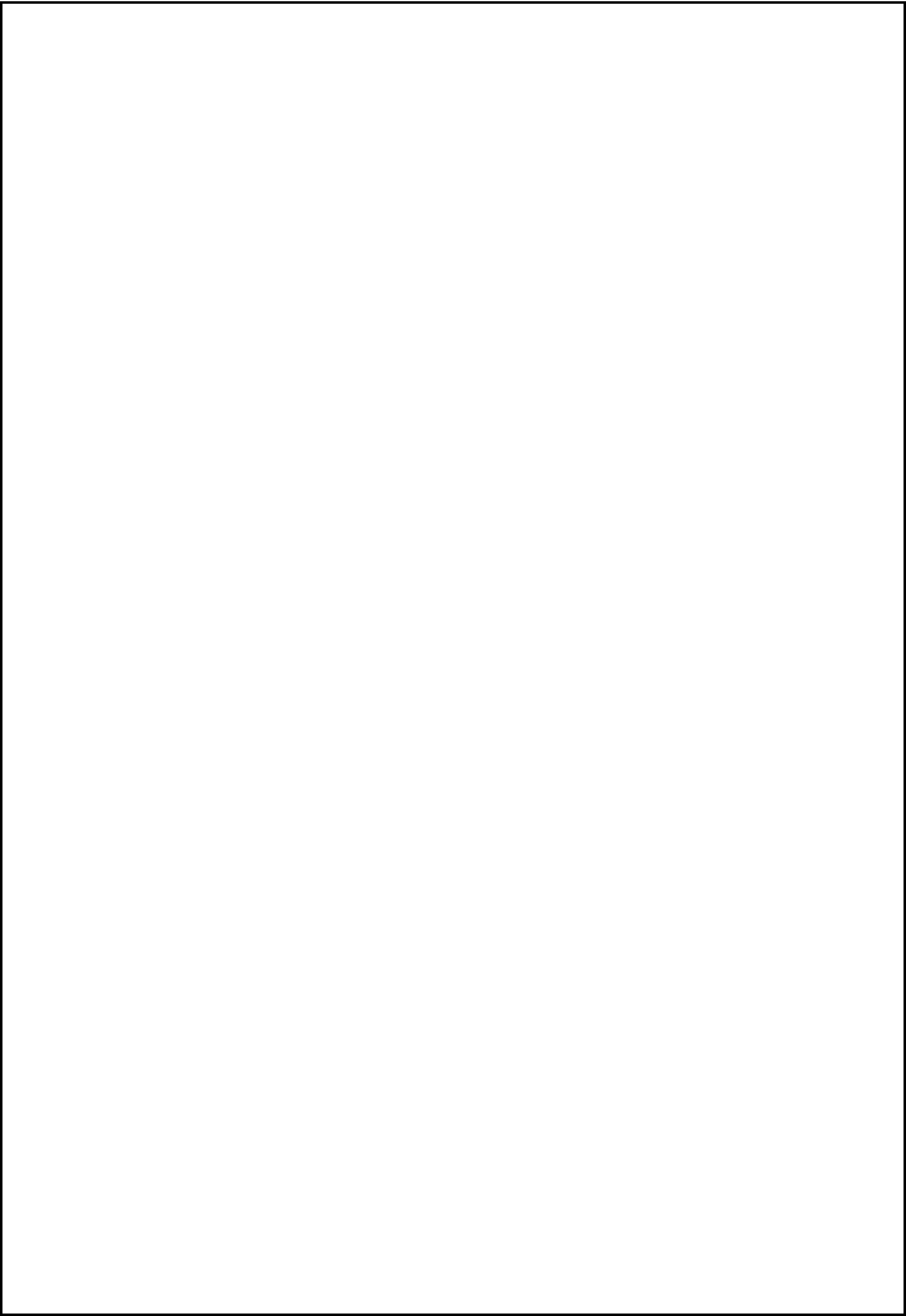




CHAPTER 1

SMART LIBRARY REQUEST WORKFLOW USING SERVICENOW

Executive Summary

The **Smart Library Request Workflow** is a ServiceNow-based application developed to automate and simplify the library book borrowing process. Traditional library systems often rely on manual procedures for issuing books, maintaining records, approving requests, and tracking borrowed books. These manual operations consume significant time, increase paperwork, and may lead to human errors such as duplicate requests, inaccurate book status updates, and inefficient record management.

This project addresses these challenges by developing a workflow-driven library management solution using the ServiceNow platform. The application enables students to submit borrow requests digitally while librarians manage book approvals through an automated workflow. The implementation uses several core ServiceNow platform capabilities, including **ServiceNow Studio, Flow Designer, Access Control Lists (ACLs), UI Policies, Reference Qualifiers, Email Notifications, and Reporting.**

When a student submits a borrow request, the system automatically initiates an approval process for the librarian. After approval, the workflow updates the availability status of the selected book, modifies the borrow request status, and sends an email notification to the requester. The system also prevents students from requesting books that are already issued by applying a reference qualifier that displays only available books. Furthermore, a UI Policy improves data consistency by making the Return Date mandatory whenever a book is issued.

To provide analytical insights, the project includes a **Most Borrowed Books** report, available in both bar chart and pie chart formats. These reports help librarians identify frequently borrowed books and understand borrowing trends, supporting better inventory management and decision-making.

Role-based security has been implemented using Access Control Lists (ACLs), ensuring that students and librarians have access only to the functionalities relevant to their responsibilities. Students can create and track borrow requests, while librarians can manage books, approve requests, and maintain library records securely.

Overall, the Smart Library Request Workflow demonstrates how ServiceNow's low-code platform can automate business processes, improve operational efficiency, enhance data security, and provide a scalable solution for library management. The project serves as a practical example of workflow automation and enterprise application development using the ServiceNow platform.

CHAPTER 2

Project Overview

Introduction

Library management is an essential function in educational institutions, organizations, and public libraries. As the number of books and users increases, maintaining borrowing records manually becomes increasingly difficult. Traditional systems require librarians to manually verify book availability, process borrow requests, update records, and maintain transaction histories. These manual procedures consume time, increase the likelihood of errors, and reduce operational efficiency.

Modern organizations are increasingly adopting workflow automation platforms to streamline such repetitive processes. ServiceNow provides a powerful low-code development environment that enables organizations to automate workflows, manage approvals, enforce business rules, and improve operational efficiency through configurable applications.

The **Smart Library Request Workflow** is a custom application developed using the ServiceNow platform to automate the complete library borrowing process. The application replaces manual operations with an intelligent workflow that allows students to request books digitally while enabling librarians to approve, reject, and manage borrow requests efficiently.

The application begins with students selecting an available book and submitting a borrow request. The system automatically routes the request to the librarian for approval using Flow Designer. Upon approval, the workflow updates both the Book and Borrow Request records, changes the availability status of the book, sends an email notification to the requester, and records the transaction for reporting purposes.

Security is implemented using Access Control Lists (ACLs), ensuring that users can only perform operations permitted by their assigned roles. Reference Qualifiers prevent students from requesting books that are already issued, while UI Policies improve data consistency by enforcing mandatory fields during specific workflow stages.

The project also includes reporting capabilities that allow librarians to analyze borrowing trends through the **Most Borrowed Books** report. This enables better decision-making regarding library resources and inventory planning.

The Smart Library Request Workflow demonstrates how ServiceNow can be used to build enterprise-grade workflow applications with minimal coding while maintaining security, scalability, and maintainability.

Existing System

In many libraries, book borrowing is still managed through manual or semi-automated processes. Students submit requests physically or through basic record-keeping systems, and librarians manually verify book availability before issuing books. Such systems have several limitations:

- Manual approval process increases processing time.
- Book availability is difficult to track accurately.
- Duplicate borrowing requests may occur.
- Record management requires significant manual effort.

- Limited reporting capabilities.
- Poor transparency for students regarding request status.
- Higher possibility of human errors.
- Difficult to maintain historical borrowing records.

These limitations reduce the efficiency of library operations and increase administrative workload.

Proposed System

The proposed Smart Library Request Workflow automates the complete borrowing lifecycle using ServiceNow.

The system provides:

- Digital borrow request submission.
- Automated approval workflow.
- Automatic book status updates.
- Borrow request status tracking.
- Email notifications.
- Role-based security.
- Dynamic form behavior using UI Policies.
- Intelligent filtering using Reference Qualifiers.
- Analytical reports for library management.

The application significantly reduces manual work while improving efficiency, transparency, and user experience.

CHAPTER 3

Problem Statement

Libraries play an essential role in providing academic resources to students and faculty. However, many library operations are still performed manually, making the borrowing process time-consuming and difficult to manage. Students often depend on librarians to manually verify book availability, record borrowing details, approve requests, and update book records. As the number of users and books increases, maintaining accurate records becomes increasingly challenging.

Traditional library systems face several operational issues, including delays in processing requests, duplicate borrowing attempts, inaccurate tracking of book availability, and inefficient management of borrowing records. Manual data entry also increases the possibility of human errors, resulting in inconsistencies in the library database.

Another major challenge is the lack of workflow automation. Borrow requests require manual approval, and there is no centralized mechanism to notify students about the status of their requests. Library administrators also face difficulties in generating analytical reports to understand borrowing trends and optimize book inventory.

To address these challenges, an automated workflow-based solution is required. The Smart Library Request Workflow developed using ServiceNow automates the complete borrowing process through workflow automation, role-based access control, UI policies, reference qualifiers, email notifications, and reporting. The system improves operational efficiency, reduces manual intervention, and enhances the overall user experience for both students and librarians.

Objectives

The primary objective of this project is to design and implement an automated library management workflow using the ServiceNow platform that simplifies book borrowing and improves the efficiency of library operations.

The specific objectives of the project are:

- To develop a digital platform for managing library book borrowing requests.
- To automate the approval process using ServiceNow Flow Designer.
- To allow students to submit book borrowing requests online.
- To enable librarians to approve or reject requests efficiently.
- To automatically update the availability status of books after approval.
- To implement role-based security using Access Control Lists (ACLs).
- To improve data accuracy through UI Policies and Reference Qualifiers.
- To send automated email notifications to users regarding request status.
- To generate reports for analyzing borrowing trends.
- To provide a scalable and maintainable solution using ServiceNow's low-code platform.

Scope of the Project

The Smart Library Request Workflow focuses on automating the core activities involved in managing library book borrowing within an organization or educational institution. The project provides an efficient workflow for students to request books and for librarians to manage approvals and book availability.

The scope of the project includes the following functionalities:

Book Management

The application allows librarians to maintain book records, including title, author, ISBN, category, and availability status. The availability of books is automatically updated when books are issued.

Borrow Request Management

Students can submit borrowing requests for available books. The application stores all borrowing transactions in a dedicated Borrow Request table for future reference.

Workflow Automation

ServiceNow Flow Designer automates the approval process by routing requests to librarians, updating records, and sending notifications based on approval decisions.

User Role Management

Different user roles are implemented to ensure secure access to the application. Students can create borrow requests, while librarians are responsible for approving requests and managing books.

Security

Access Control Lists (ACLs) ensure that users only access the information and operations relevant to their assigned roles.

Data Validation

Reference Qualifiers ensure that students can only request books that are currently available. UI Policies improve form usability by enforcing mandatory fields during specific workflow stages.

Reporting

The application generates analytical reports such as "Most Borrowed Books," helping librarians understand borrowing patterns and improve inventory planning.

CHAPTER 4

System Requirements

Hardware Requirements

The Smart Library Request Workflow is a cloud-based application developed on the ServiceNow platform. Since the application is hosted on ServiceNow servers, users require only a standard computing device with internet connectivity to access the system.

The minimum hardware requirements are:

Component	Specification
Processor	Intel Core i3 or Higher
RAM	Minimum 4 GB (8 GB Recommended)
Storage	20 GB Free Disk Space
Internet	Broadband Connection
Display	1366 × 768 Resolution or Higher

Software Requirements

The software requirements for developing and running the application are as follows:

Software	Purpose
ServiceNow Personal Developer Instance (PDI)	Application Development
ServiceNow Studio	Building the Application
Flow Designer	Workflow Automation
Google Chrome / Microsoft Edge	Web Browser
Windows 10/11	Operating System
Git & GitHub	Version Control
Microsoft Word	Documentation
Google Drive	Demo Video Sharing

CHAPTER 5

Technologies Used

The Smart Library Request Workflow was developed using the ServiceNow platform, which provides a low-code development environment for building enterprise applications. Several built-in ServiceNow components were utilized to automate workflows, enforce security, manage records, and generate reports. The following technologies were implemented during the development of the project.

5.1 ServiceNow Studio

ServiceNow Studio is the primary development environment used to build custom applications on the ServiceNow platform. It provides an integrated workspace where developers can create tables, forms, modules, business rules, workflows, UI policies, and other application components without extensive programming.

In this project, ServiceNow Studio was used to create the **Smart Library Request Workflow** application. The custom Book and Borrow Request tables, application modules, and other configurations were developed within the Studio environment. Studio also helped organize all application files and provided version management throughout the development process.

Key Features Used

- Custom Application Development
- Custom Tables
- Form Design
- Module Creation
- Application File Management

5.2 Flow Designer

Flow Designer is ServiceNow's low-code workflow automation tool. It enables developers to automate business processes by creating workflows using drag-and-drop actions instead of writing complex scripts.

The Flow Designer is the core component of this project. It automates the entire library borrowing process by initiating an approval request whenever a student submits a borrow request. Based on the librarian's decision, the flow updates the status of both the Book and Borrow Request records and sends an email notification to the student.

The workflow implemented in this project includes the following steps:

- Trigger when a Borrow Request is created.
- Send approval request to the Librarian.
- Wait for approval.
- Update Book Status.
- Update Borrow Request Status.
- Send Email Notification.

This automation significantly reduces manual effort and ensures consistency throughout the borrowing process.

5.3 Access Control Lists (ACL)

Access Control Lists (ACLs) are ServiceNow security mechanisms used to control access to records and operations within the application. ACLs define which users can create, read, update, or delete specific records based on their assigned roles.

In the Smart Library Request Workflow, ACLs were configured for both the Book and Borrow Request tables to ensure secure access to application data.

Students are permitted to create borrow requests and access only the information relevant to them, while librarians have additional privileges to manage books and process approval requests. This implementation enhances application security and prevents unauthorized access.

5.4 User Roles

Role-Based Access Control (RBAC) is implemented using ServiceNow roles. Different roles provide different levels of access according to user responsibilities.

The application includes the following user roles:

Student

The Student role allows users to:

- View available books.
- Submit borrow requests.
- Track request status.

Librarian

The Librarian role is responsible for:

- Managing book records.
- Reviewing borrow requests.
- Approving or rejecting requests.
- Monitoring reports.

Administrator

The Administrator role manages:

- Application configuration.
- User management.
- Security settings.
- System maintenance.

Using separate roles ensures that each user accesses only the features necessary for their responsibilities.

5.5 UI Policy

UI Policies are used to dynamically control the behavior of forms without writing scripts. They can make fields mandatory, read-only, or visible based on specific conditions.

In this project, a UI Policy was implemented to improve data accuracy. When the status of a borrowed book changes to **Issued**, the Return Date field becomes mandatory. This ensures that all issued books have an associated return date, improving record management and reducing incomplete data.

The UI Policy provides a better user experience by automatically enforcing business rules during data entry.

5.6 Reference Qualifier

Reference Qualifiers are used to filter records displayed in reference fields. They ensure that users can only select valid records according to predefined conditions.

The Borrow Request form contains a reference field for selecting books. A Reference Qualifier was configured to display only books whose availability status is **Available**. Books that have already been issued are automatically excluded from the selection list.

This feature prevents duplicate borrowing requests and improves the reliability of the borrowing process.

5.7 Email Notification

Email Notifications provide automated communication between the system and users. Notifications keep users informed about important events without requiring manual intervention.

In this project, an email notification is automatically generated after a librarian approves a borrow request. The notification informs the student about the approval and confirms that the requested book has been successfully issued.

Automated notifications improve communication and reduce the need for manual follow-up.

5.8 Reports

ServiceNow Reports provide graphical representations of application data that help users analyze trends and monitor system performance.

The Smart Library Request Workflow includes the **Most Borrowed Books** report in both **Bar Chart** and **Pie Chart** formats. These reports summarize borrowing activity by counting the number of borrow requests for each book.

The reports assist librarians in identifying frequently borrowed books and support better library inventory planning.

5.9 GitHub

GitHub was used to maintain the project repository and documentation. The repository includes screenshots, project reports, architecture diagrams, demo links, and a detailed README file describing the application's functionality and implementation.

Version control through GitHub allows easy project sharing, documentation, and future enhancements.

5.10 Google Drive

Google Drive was used to store and share demonstration videos of the project. The repository includes a Demo Link file that directs users to the Google Drive folder containing the complete workflow demonstrations.

The demo videos illustrate:

- Borrow Request Creation
- Approval Workflow
- Complete Smart Library Request Workflow

Roles ☆			
Search			
Actions on selected rows...			
New			
All			
☐	Name ▲	Description	Elevated privilege
☐	web_service_configuration_admin	Can manage web service configurations that control behavior and security. This includes REST, GraphQL, and SOAP.	false
☐	wizard_admin	Can manage the Catalog Wizard, including Wizard steps, Wizard sections, Wizard questions and Wizard feedback.	false
☐	workflow_admin	Can create, edit, publish or delete graphical workflows	false
☐	workflow_creator	Can create new graphical workflows	false
☐	workflow_publisher	Can publish graphical workflows	false
☐	workflow_report_viewer	Access to workflow scratchpad for reports	false
☐	workspace.activity_stream_restrict_creat...		false
☐	workspace_admin		false
☐	workspace_configure_form_field	Ability to do field level configuration in workspace form	false
☐	workspace_list_admin	Can manage workspace lists and categories	true
☐	workspace_user		false
☐	x_2120474_smart_0.admin	Default admin role	false
☐	x_2120474_smart_0.librarian	Librarian role to manage books and approve requests.	false
☐	x_2120474_smart_0.student	Student role to request and track library books.	false

User - student user ☆			
Search			
Actions on selected rows...			
Edit...			
User student user			
User ID	student1	Email	student1@example.com
First name	student	Language	-- None --
Last name	user	Calendar integration	Outlook
Title		Time zone	System (America/Los_Angeles)
Department		Date format	System (2026-08-19)
Password needs reset	☐	Business phone	
Locked out	☐	Mobile phone	
Active	☒	Photo	Click to add...
Identity type	Human		
Internal Integration User	☐		
Update Set Password Delete			
Related Links			
View linked accounts			
View Subscriptions			
Reset a password			
Roles (2) Groups (1) Delegates User Client Certificates			
Role Search			
Actions on selected rows...			
Edit...			
User = student user			
☐	Role	State	Inheritance Count
☐	snc_required_script_writer_permission	Active	1
☐	x_2120474_smart_0.student	Active	1

FavoritesHistoryWorkspacesAdmin

User - Librarian user

Search

User Librarian user

UpdateSet PasswordDelete

User IDlibrarian1

First nameLibrarian

Last nameuser

Title

Department

Password needs reset

Locked out

Active

Identity typeHuman

Internal Integration User

Emaillibrarian1@example.com

Language-- None --

Calendar integrationOutlook

Time zoneSystem (America/Los_Angeles)

Date formatSystem (2026-08-19)

Business phone

Mobile phone

PhotoClick to add...

UpdateSet PasswordDelete

Related Links

[View linked accounts](#)

[View Subscriptions](#)

[Reset a password](#)

Roles (2)Groups (1)DelegatesUser Client Certificates

RoleSearch

Actions on selected rows...Edit...

User = Librarian user

Role	State	Inherited	Inheritance Count
x_2120474_smart_0.librarian	Active	false	
snc_required_script_writer_permission	Active	true	1

Borrow Request Approval FlowInactive

View:TestTest Activate

TRIGGER

Borrow Request Created or Updated where (Status is Requested)

TriggerCreated or Updated

* TableBorrow Request [x_2120474_sm...]

ConditionStatus is Requested

Run TriggerOnce

Advanced Options

DeleteCancelDone

1Ask For Approval on Borrow Request

Action Properties

ActionAsk For Approval

Action Inputs

* RecordTrigger - ... Borrow Request...

TableBorrow Request [x_2120474_sm...]

Approval Reason

Approval FieldStatus

Journal FieldSelect a field

* RulesApproveWhenAnyone approvesLibrarian user X

Due DateNone

Add another OR rule set

DeleteCancelDone

DataCollapse All

Flow Variables

Trigger - Record Created or Updated

Borrow Request RecordRecord

Changed FieldsArray.Object

Borrow Request TableTable

Run Start Time UTCDate/Time

Run Start Date/TimeDate/Time

1 - Ask For Approval

Approval StateChoice

Action StatusObject

Status: ModifiedApplication: Smart Library Request Workflow

0

Workflow Studio

Borrow Request Approval Flow

Workflow Studio

Borrow Request Approval Flow

Flow • Smart Library Request Workflow

Active

Deactivate

Activate

TRIGGER

Click or hit enter to Expand ServiceNow Core Record Trigger for Borrow Request Created or Updated where (Status is Requested)

Borrow Request Created or Updated where (Status is Requested)

ACTIONS

Select multiple

1

Ask For Approval on Borrow Request

2

Update Book Record

3

Update Borrow Request Record

4

Send Email

+

Add an Action, Flow Logic, or Subflow

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Data

Collapse All

Flow Variables

Trigger - Record Created or Updated

Borrow Request Record

Record

Changed Fields

Array/Object

Borrow Request Table

Table

Run Start Time UTC

Date/Time

Run Start Date/Time

Date/Time

1 - Ask For Approval

2 - Update Record

3 - Update Record

4 - Send Email

email

Record

Action Status

Object

Favorites

History

Workspaces

Admin

UI Policy - Make Return Date mandatory when Issued

Search

Update

Delete

UI Policy

Make Return Date mandatory when Issued

Update

Delete

UI policies change fields on a form based on a set of conditions. Use UI policies to show or hide fields, or to make fields mandatory or read-only based on these conditions. A UI policy specifies one or more actions to take when the policy is triggered. First, create the policy. Then add as many actions as necessary. UI policy actions are applied only if all of the following conditions are met:

- The UI Policy is Active
- The items in the Conditions field evaluate to true
- The field specified in the UI policy action is present on the specified form

[More info](#)

Table

Borrow Request [x_2120474_smart_0_bo...

Application

Smart Library Request Workflow

Active

* Short description

Make Return Date mandatory when Issued

When to Apply

Conditions

Add Filter Condition

Add OR Clause

Status

is

Issued

AND

OR

Update

Delete

Related Links

Advanced view

Run Point Scan

UI Policy Actions (1)

UI Policy Related List Actions

for text

Search

Actions on selected rows...

New

UI policy = Make Return Date mandatory when Issued

Field name	Mandatory	Visible	Read only
return_date	True	True	Leave alone

1 to 1 of 1

[More Info](#)

Application Smart Library Request Workflow ⓘ

Active ☒

Function field ☐

Mandatory ☐

Display ☐

Lead only option -- None -- ▼

Default Value

The **Reference** field specifies what table this field displays values from.

* Reference

Reference qual condition

Table Subscription Configuration (1)

≡ 🔍 Name Search ⚙️ — Actions on selected rows... ▼

Access Controls

Name ▲

Decision Type

Operation

Type

Active

Updated by

Updated

x_2120474_smart_0_book

Allow If

read

record

true

admin

2026-07-03 10:38:41

x_2120474_smart_0_book

Allow If

create

record

true

admin

2026-07-03 10:38:40

x_2120474_smart_0_book

Allow If

read

record

true

admin

2026-07-03 10:38:41

x_2120474_smart_0_book

Allow If

read

record

true

admin

2026-07-03 10:38:41

x_2120474_smart_0_book

Allow If

delete

record

true

admin

2026-07-03 10:38:42

x_2120474_smart_0_book

Allow If

delete

record

true

admin

2026-07-03 10:38:42

x_2120474_smart_0_book

Allow If

write

record

true

admin

2026-07-03 10:38:42

x_2120474_smart_0_book

Allow If

create

record

true

admin

2026-07-03 10:38:40

x_2120474_smart_0_book

Allow If

write

record

true

admin

2026-07-03 10:38:42

<<

<

1

to 9 of 9

>

>>

Table Subscription Configuration (1)

Name

Search

Actions on selected rows...

Access Controls

🔍

Name ▲

Decision Type

Operation

Type

Active

Updated by

Updated

x_2120474_smart_0_borrow_request	Allow If	create	record	true	admin	2026-07-03 13:45:06
x_2120474_smart_0_borrow_request	Allow If	read	record	true	admin	2026-07-03 13:45:06
x_2120474_smart_0_borrow_request	Allow If	delete	record	true	admin	2026-07-03 13:45:07
x_2120474_smart_0_borrow_request	Allow If	write	record	true	admin	2026-07-03 13:45:07

<<

<

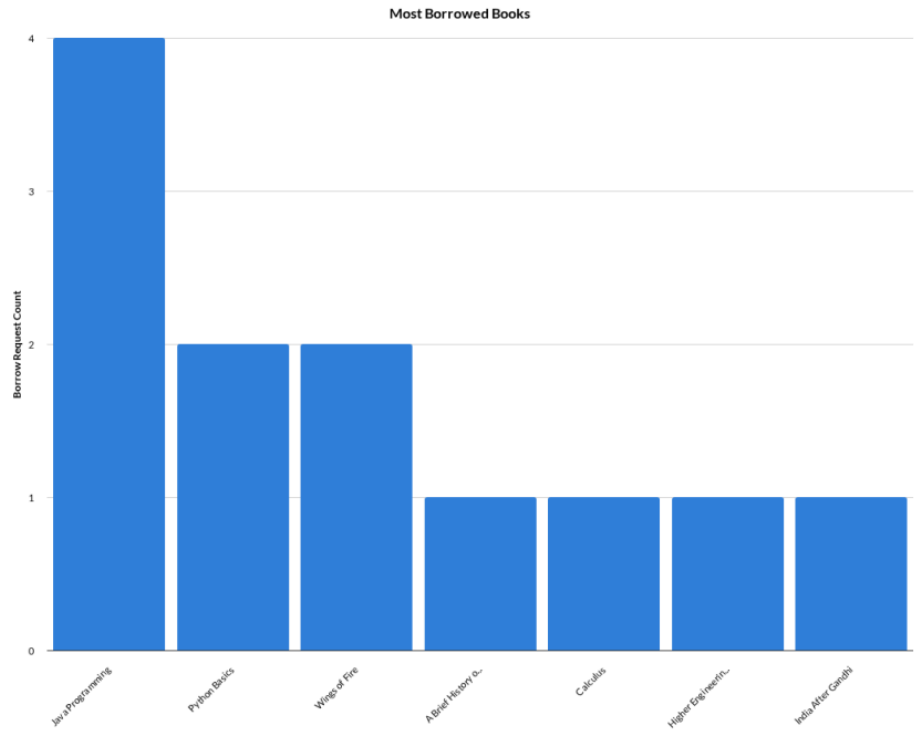
1

>

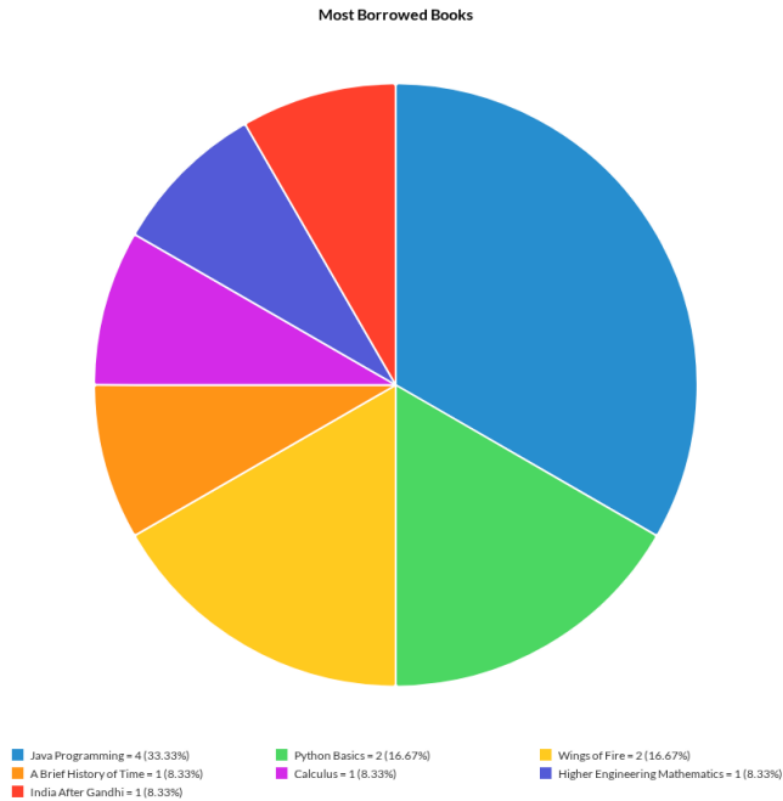
>>

to 4 of 4

Report Title: Most Borrowed Books
Run Date and Time: 2026-07-10 07:12:40 Pacific Daylight Time
Run by: System Administrator
Table name: x_2120474_smart_0_borrow_request
Group by: Book



Report Title: Most Borrowed Books
Run Date and Time: 2026-07-10 07:13:20 Pacific Daylight Time
Run by: System Administrator
Table name: x_2120474_smart_0_borrow_request
Group by: Book



CHAPTER 6

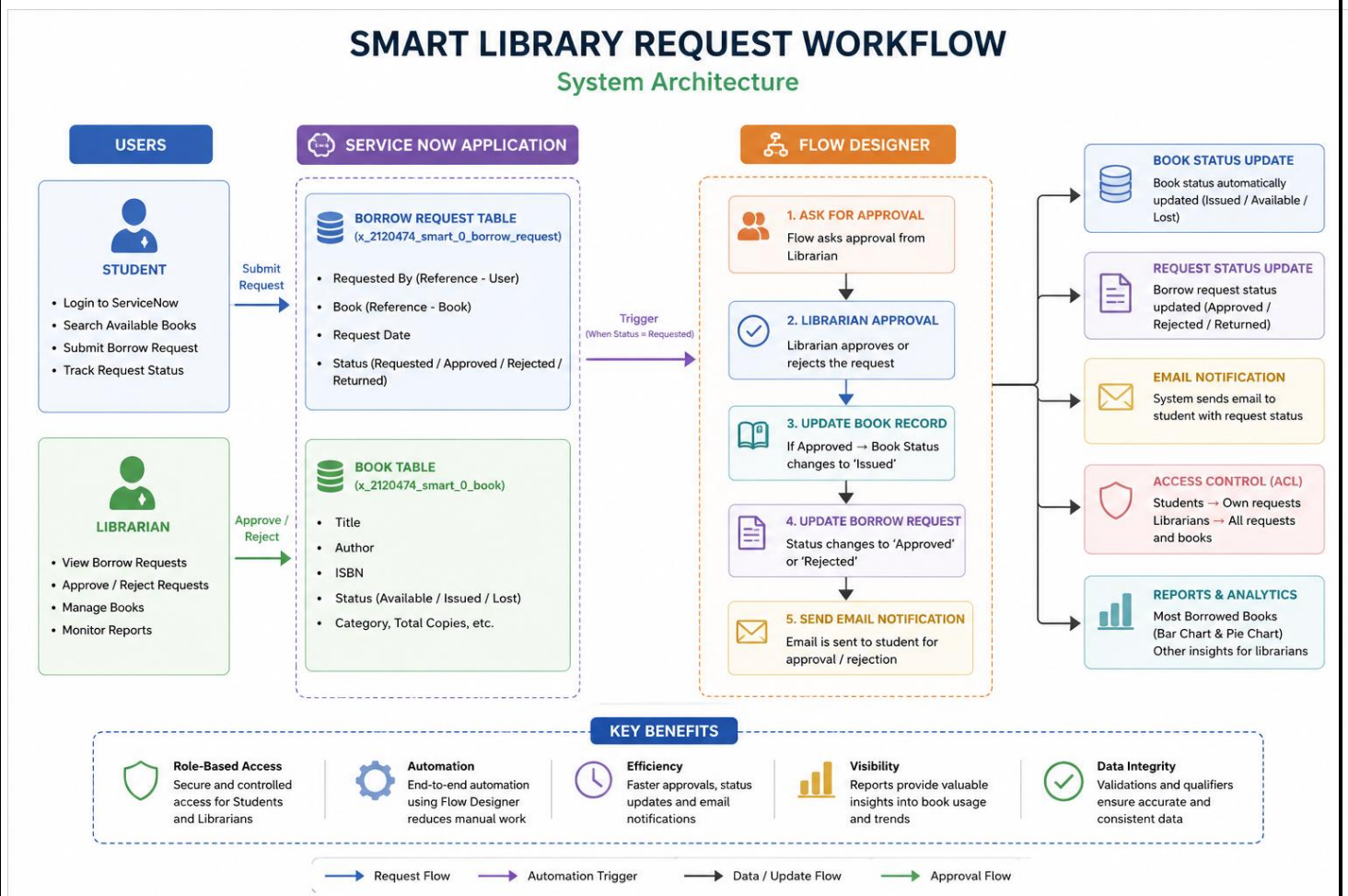
System Architecture and Application Design

6.1 System Architecture

The Smart Library Request Workflow follows a workflow-driven architecture designed using the ServiceNow platform. The architecture integrates multiple ServiceNow components, including custom tables, Flow Designer, Access Control Lists (ACLs), UI Policies, Reference Qualifiers, Reports, and Email Notifications, to automate the complete library borrowing process.

The architecture begins when a student submits a borrow request through the application interface. The request is stored in the Borrow Request table and automatically triggers the Flow Designer workflow. The workflow sends an approval request to the librarian, who reviews and either approves or rejects the request. If approved, the system updates the Book table to mark the selected book as **Issued**, updates the Borrow Request record, and sends an email notification to the student. The reporting module then analyzes the borrowing data to generate insights such as the "Most Borrowed Books" report.

The architecture is designed to ensure secure data access, automated processing, and efficient management of library resources.



6.2 Architecture Components

The Smart Library Request Workflow consists of the following major components:

Student

The Student is the primary user of the application. Students can browse available books and submit borrow requests. They can also track the status of their requests after submission.

Responsibilities

- View available books.
- Submit borrow requests.
- Receive email notifications.
- Track request status.

Librarian

The Librarian manages the borrowing process by reviewing and approving or rejecting requests. The librarian is also responsible for maintaining book records and monitoring reports.

Responsibilities

- Approve borrow requests.
- Reject borrow requests.
- Update book records.
- Monitor reports.
- Manage library inventory.

Book Table

The Book table stores all information related to books available in the library.

Each record contains information such as:

- Book ID
- Title
- Author
- Category
- ISBN
- Availability Status

Whenever a borrow request is approved, the Book Status is automatically updated by the workflow.

Borrow Request Table

The Borrow Request table stores all book borrowing transactions.

Each request includes:

- Requested By
- Selected Book
- Request Date
- Status

This table serves as the central component for tracking all borrowing activities.

Flow Designer

Flow Designer automates the borrowing process without requiring custom scripting.

The implemented flow performs the following actions:

1. Trigger when a Borrow Request is created.
2. Ask for Librarian Approval.
3. Update Book Status.
4. Update Borrow Request Status.
5. Send Email Notification.

This automation reduces manual intervention and ensures process consistency.

Access Control Lists (ACL)

ACLs protect application data by controlling access to records based on user roles.

The application implements ACLs for:

- Book Table
- Borrow Request Table

These ACLs ensure that users can only perform operations permitted by their assigned roles.

UI Policy

The UI Policy improves data quality by dynamically modifying form behavior.

In this project, the Return Date field becomes mandatory whenever a book is issued.

This prevents incomplete borrowing records.

Reference Qualifier

The Reference Qualifier filters the list of books displayed in the Borrow Request form.

Only books whose status is **Available** are displayed to students.

Books already marked as **Issued** are automatically excluded.

This prevents duplicate borrowing requests.

Email Notification

Email Notifications automatically inform students about the approval of their borrow requests.

The notification includes confirmation that the requested book has been successfully approved and issued.

Reports

The reporting component analyzes borrowing transactions stored in the Borrow Request table.

The project includes:

- Most Borrowed Books (Bar Chart)
- Most Borrowed Books (Pie Chart)

These reports assist librarians in understanding borrowing patterns and planning library resources.

6.3 Overall Workflow

The Smart Library Request Workflow follows the sequence below:

Student



Borrow Request Form



Borrow Request Table



Flow Designer Trigger



Ask for Approval



Librarian Approval



Update Book Status



Update Borrow Request Status



Send Email Notification



Reports & Analytics

This workflow ensures that every borrow request is processed systematically, securely, and efficiently.

6.4 Application Navigation

The Smart Library Request Workflow application contains several modules that allow users to perform specific tasks.

Book Management

This module enables librarians to manage book records by adding, updating, and maintaining book information.

Borrow Request Management

Students can create borrow requests, while librarians review and manage submitted requests.

Reports

The Reports module displays graphical representations of borrowing data, helping librarians monitor library usage.

User Management

Different user roles such as Student, Librarian, and Administrator are managed through ServiceNow's role-based access control mechanism.

Table Book

DeleteUpdateDelete All Records

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* LabelBook

ApplicationSmart Library Request Workflow

* Namex_2120474_smart_0_book

Remote Table

ColumnsControlsApplication Access

Table Columns

for text

Search

1 to 13 of 13

New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Updated	Date/Time	(empty)		40	false
Author	String	(empty)		100	false
Book ID	String	(empty)		50	false
Created by	String	(empty)		40	false
ISBN	String	(empty)		40	false
Availability Status	Choice	(empty)		40	false
Title	String	(empty)		200	false
Total Copies	Integer	(empty)		40	false
Created	Date/Time	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false
Category	Choice	(empty)		40	false
Insert a new row...					

Table - Borrow Request

Search

Table Borrow Request

DeleteUpdateDelete All Records

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* LabelBorrow Request

ApplicationSmart Library Request Workflow

* Namex_2120474_smart_0_borrow_request

Remote Table

ColumnsControlsApplication Access

Table Columns

for text

Search

1 to 10 of 10

New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Book	Reference	Book		32	false
Updated	Date/Time	(empty)		40	false
Request Date	Date/Time	(empty)		40	false
Status	Choice	(empty)		40 Requested	false
Created by	String	(empty)		40	false
Created	Date/Time	(empty)		40	false
Requested By	Reference	User [sys_user]		32	false
Sys ID	Sys ID (GUID)	(empty)		32	false
Updated by	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Insert a new row...					

DeleteUpdateDelete All Records

FavoritesHistoryWorkspacesAdmin

Books

Search

Remove favorite for Books

Actions on selected rows...

New

All

Book ID	Title	Author	Category	ISBN	Availability Status	Total Copies
Search	Search	Search	Search	Search	Search	Search
B007	The Alchemist	Paulo Coelho	Novel	9780061122415	Available	6
B004	The Selfish Gene	Richard Dawkins	Science	9780198788607	Available	2
B001	Java Programming	James Gosling	Programming	9781234567890	Available	5
B006	Sapiens	Yuval Noah Harari	History	9780062316097	Available	4
B011	Higher Engineering Mathematics	B. S. Grewal	Mathematics	9788193328491	Available	5
B010	Electrical Engineering Fundamentals	Vincent Del Toro	Engineering	9780133760031	Available	2
B005	India After Gandhi	Ramachandra Guha	History	9780330396110	Available	3
B012	Calculus	James Stewart	Mathematics	9781285741550	Available	4
B009	Basic Civil Engineering	S. S. Bhavikatti	Engineering	9788121926751	Available	3
B003	A Brief History of Time	Stephen Hawking	Science	9780553380163	Available	4
B008	Wings of Fire	Dr. A. P. J. Abdul Kalam	Novel	9788173711466	Available	5
B002	Python Basics	Guido van Rossum	Programming	9789876543210	Available	3

1 to 12 of 12

FavoritesHistoryWorkspacesAdmin

Borrow Request - Created 2026-07-08 09:17:45

Search

UpdateDelete

Borrow Request
Created 2026-07-08 09:17:45

Requested BySystem Administrator

Request Date2026-07-08 09:17:28

BookJava Programming

Statusrequested

UpdateDelete

FavoritesHistoryWorkspacesAdmin

Books

Search

Actions on selected rows...

New

All

Book ID	Title	Author	Category	ISBN	Availability Status	Total Copies
Search	Search	Search	Search	Search	Search	Search
B007	The Alchemist	Paulo Coelho	Novel	9780061122415	Available	6
B004	The Selfish Gene	Richard Dawkins	Science	9780198788607	Available	2
B001	Java Programming	James Gosling	Programming	JPO01	Issued	5
B006	Sapiens	Yuval Noah Harari	History	9780062316097	Available	4
B011	Higher Engineering Mathematics	B. S. Grewal	Mathematics	9788193328491	Issued	5
B010	Electrical Engineering Fundamentals	Vincent Del Toro	Engineering	9780133760031	Available	2
B005	India After Gandhi	Ramachandra Guha	History	9780330396110	Issued	3
B012	Calculus	James Stewart	Mathematics	9781285741550	Issued	4
B009	Basic Civil Engineering	S. S. Bhavikatti	Engineering	9788121926751	Available	3
B003	A Brief History of Time	Stephen Hawking	Science	9780553380163	Issued	4
B008	Wings of Fire	Dr. A. P. J. Abdul Kalam	Novel	9788173711466	Issued	5
B002	Python Basics	Guido van Rossum	Programming	9789876543210	Issued	3

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CHAPTER 7

Database Design

7.1 Introduction

Database design is one of the most important phases in application development. It determines how information is stored, maintained, and retrieved throughout the application's lifecycle. A well-designed database ensures efficient data management, minimizes redundancy, and maintains data integrity.

The Smart Library Request Workflow application uses two custom tables developed within the ServiceNow platform:

- **Book Table**
- **Borrow Request Table**

These tables work together to manage library books and borrowing transactions.

The Book table stores the details of all books available in the library, while the Borrow Request table records every borrowing transaction performed by students. The relationship between these tables enables efficient tracking of issued books and borrowing history.

7.2 Book Table

The Book table is the master table of the application. It contains information about every book available in the library.

Each record in this table represents one physical book available for borrowing.

The table enables librarians to maintain the complete inventory of books and monitor their availability.

Purpose of the Book Table

The Book table is responsible for:

- Maintaining book records.
- Tracking book availability.
- Storing book information.
- Preventing duplicate borrowing.
- Supporting report generation.

Book Table Fields

Field Name	Data Type	Description
Book ID	String	Unique identifier of the book
Title	String	Name of the book
Author	String	Author of the book
Category	Choice	Book category
ISBN	String	International Standard Book Number
Status	Choice	Availability status of the book

Explanation of Fields

Book ID

Book ID uniquely identifies every book stored in the library.

Each book receives a unique identifier, making it easier to search and manage records.

Title

The Title field stores the complete name of the book.

Example:

- Java Programming
- Python Basics
- Database Management Systems

Author

This field stores the author's name.

Examples:

- James Gosling
- Guido van Rossum
- Robert Lafore

Category

The Category field classifies books into different subject areas.

Examples:

- Programming
- Database
- Networking
- Operating Systems
- Artificial Intelligence

This helps students quickly identify books belonging to a particular domain.

ISBN

ISBN uniquely identifies a published book.

It ensures accurate identification of books and avoids duplication.

Status

The Status field indicates whether a book is currently available.

Possible values include:

- Available
- Issued
- Lost

Initially, all books are marked as **Available**.

When a borrow request is approved, Flow Designer automatically updates the status to **Issued**.

7.3 Borrow Request Table

The Borrow Request table records every borrowing transaction initiated by students.

Whenever a student requests a book, a new record is created in this table.

The Borrow Request table serves as the central workflow table because Flow Designer is triggered whenever a new request is submitted.

Purpose

The Borrow Request table is responsible for:

- Recording borrow requests.
- Managing request status.
- Tracking borrowing history.
- Triggering workflow automation.
- Supporting report generation.

Borrow Request Fields

Field Name	Data Type	Description
Requested By	Reference	Student requesting the book
Book	Reference	Selected book
Request Date	Date/Time	Date of request
Status	Choice	Current request status

Explanation of Fields

Requested By

This field stores the student who submits the borrow request.

It references the **User** table.

Book

This field references the Book table.

Instead of storing the book name directly, it stores a reference to the corresponding Book record.

The Reference Qualifier ensures that only books whose Status is **Available** appear in this field.

Request Date

The Request Date stores the exact date and time when the student submitted the request.

This information helps maintain borrowing history.

Status

The Status field indicates the current stage of the borrowing process.

Possible values include:

- Requested
- Approved
- Rejected
- Returned

Initially, every new request is marked as **Requested**.

After librarian approval, Flow Designer automatically updates the status to **Approved**.

7.4 Relationship Between Tables

The Book table and Borrow Request table are connected through a **Reference Field**.

The Borrow Request table contains a Book field that references the Book table.

Relationship:

Book Table

↓ Reference

Borrow Request Table

This relationship enables:

- Easy retrieval of book details.
- Efficient workflow automation.
- Report generation.
- Data consistency.
- Prevention of duplicate records.

7.5 Database Normalization

The database is designed following normalization principles to reduce redundancy.

The Book information is stored only once in the Book table.

Borrow Request records simply reference the corresponding Book record instead of storing duplicate book information.

Advantages include:

- Reduced storage requirements.
- Improved consistency.
- Easier maintenance.
- Faster updates.
- Better scalability.

CHAPTER 8

Workflow Automation Using Flow Designer

8.1 Introduction

Workflow automation is one of the most powerful capabilities provided by the ServiceNow platform. It enables organizations to automate repetitive business processes without extensive programming by using a graphical, low-code interface. ServiceNow Flow Designer allows developers to create workflows through configurable actions, conditions, and approvals, ensuring that business processes are executed consistently and efficiently.

In the Smart Library Request Workflow project, Flow Designer automates the complete lifecycle of a book borrowing request. Instead of manually verifying requests, updating records, and notifying users, these tasks are performed automatically once a student submits a borrow request. This automation reduces manual effort, improves operational efficiency, minimizes human errors, and provides a better user experience.

The implemented workflow consists of several sequential actions, including triggering the flow, requesting librarian approval, updating book information, updating borrow request status, and sending email notifications to the student.

8.2 Flow Overview

The Smart Library Request Workflow follows the sequence illustrated below.

Student Creates Borrow Request



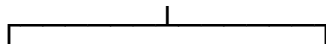
Trigger Flow



Ask for Librarian Approval



Approval Decision



Approved

Rejected



Update Book Status



Update Status



Update Borrow Request



Send Email Notification



Flow Completed

This workflow ensures that every borrow request follows the same standardized process, reducing inconsistencies and improving system reliability.

8.3 Flow Trigger

The Trigger is the starting point of the workflow. It monitors the Borrow Request table and automatically starts the workflow whenever a new borrow request is created.

Purpose

The trigger eliminates the need for manual intervention by automatically initiating the approval process immediately after a student submits a request.

Configuration

Property	Value
Trigger Type	Record Created
Table	Borrow Request
Condition	New Borrow Request Created

Working

Whenever a student submits a borrow request:

1. A new record is inserted into the Borrow Request table.
2. The trigger detects the newly created record.
3. The workflow starts automatically.
4. The Borrow Request record becomes the input for all subsequent workflow actions.

Borrow Request Approval Flow Inactive

TRIGGER

Borrow Request Created or Updated where (Status is Requested)

Trigger: Created or Updated

* Table: Borrow Request [x_2120474_sm...]

Condition: Status is Requested

Run Trigger: Once

Advanced Options

Buttons: Delete, Cancel, Done

Data Panel:

- Flow Variables
- Trigger - Record Created or Updated
 - Borrow Request Record (Record)
 - Changed Fields (Array.Object)
 - Borrow Request Table (Table)
 - Run Start Time UTC (Date/Time)
 - Run Start Date/Time (Date/Time)

8.4 Ask for Approval Action

The Ask for Approval action sends an approval request to the Librarian for review.

This step ensures that no book is issued without authorization.

Purpose

To provide controlled approval of book borrowing requests before updating the library records.

Configuration

Property	Value
Approval Type	User Approval
Approver	Librarian
Record	Borrow Request

Working

After the workflow starts:

1. The approval request is sent to the Librarian.
2. The Librarian reviews the request.
3. The Librarian either:
 - Approves the request
 - Rejects the request

The workflow then follows the appropriate path based on the approval decision.

1 Ask For Approval on Borrow Request

Action Properties

Action: Ask For Approval

Action Inputs

* Record: Trigger - ... -> Borrow Request...

Table: Borrow Request [x_2120474_sm...]

Approval Reason:

Approval Field: Status

Journal Field: Select a field

* Rules: Approve When: Anyone approves Librarian user X

Due Date: None

Add another OR rule set

OR AND

Delete Cancel Done

Data Collapse All

Flow Variables

- Trigger - Record Created or Updated
 - Borrow Request Record Record
 - Changed Fields Array.Object
 - Borrow Request Table Table
 - Run Start Time UTC Date/Time
 - Run Start Date/Time Date/Time
- 1 - Ask For Approval
 - Approval State Choice
 - Action Status Object

8.5 Update Book Record

After the librarian approves the request, the workflow automatically updates the corresponding Book record.

Purpose

To indicate that the selected book is no longer available for borrowing.

Configuration

Property	Value
Action	Update Record
Table	Book
Updated Field	Status
New Value	Issued

Working

When approval is granted:

1. The workflow locates the selected book.
2. The Book Status changes from **Available** to **Issued**.
3. The updated status prevents other students from requesting the same book.

The screenshot shows the 'Update Book Record' workflow configuration. The 'Action Properties' section has 'Action' set to 'Update Record'. The 'Action Inputs' section includes: 'Record' set to 'Trigger - Record Cr... > ... > Bo...', 'Table' set to 'Book [x_2120474_smart_0_book]', and 'Fields' set to 'Availability Status' with a value of 'Issued'. A '+ Add field value' button is also present. At the bottom right are 'Delete', 'Cancel', and 'Done' buttons. On the right side, a 'Data' panel shows a tree view with 'Flow Variables', 'Trigger - Record Created or Updated', '1 - Ask For Approval', and '2 - Update Record' expanded, showing 'Book Record' (Record), 'Book Table' (Table), and 'Action Status' (Object).

8.6 Update Borrow Request Record

Once the Book record has been updated, the Borrow Request record is also updated.

Purpose

To record the outcome of the approval process.

Configuration

Property	Value
Action	Update Record
Table	Borrow Request
Updated Field	Status
New Value	Approved

Working

The workflow:

1. Updates the borrow request status.
2. Records the approval result.
3. Maintains borrowing history for reporting and future reference.

The screenshot shows the 'Update Borrow Request Record' workflow configuration. The 'Action Properties' section has 'Action' set to 'Update Record'. The 'Action Inputs' section includes: 'Record' set to 'Trigger - ... > Borrow Request...', 'Table' set to 'Borrow Request [x_2120474_sm...', and 'Fields' set to 'Status' with a value of 'Approved'. A '+ Add field value' button is also present. At the bottom right are 'Delete', 'Cancel', and 'Done' buttons.

8.7 Send Email Notification

After updating both records, the workflow sends an email notification to the student.

Purpose

To inform the student that the borrow request has been processed successfully.

Configuration

Property	Value
Recipient	Requested By
Subject	Borrow Request Approved
Message	Book issued successfully

Working

The workflow automatically:

1. Identifies the requester.
2. Sends an email containing the approval confirmation.
3. Confirms that the requested book has been issued.

4

Send Email

Action Properties

Action Inputs

Target Record

Drag and drop record data pill

Table

Select a Table

Include Watermark

☒

* To

Trigger - Re... ▶ ... ▶ Request... X

CC

BCC

* Subject

Your Borrow Request has been Approved

↶ ↷ B I [List Icons] [Link Icon] [Code Icon]

OpenSans.Sa... 1... FF

Hello,

Your borrow request has been approved.

Please collect the book from the library.

Thank You.

Delete

Cancel

Done

FavoritesHistoryWorkspacesAdmin

Email - Your Borrow Request has been Approved

Search

EmailYour Borrow Request has been ApprovedView: Outbox

MailboxOutbox

StateReady

TargetBorrow Request: Created 2026-07-08 09:1

User

UID

Created2026-07-08 09:48:14

Deleted

Weight0

Importance

SubjectYour Borrow Request has been Approved

Recipientsadmin@example.com

Body

Content type

HeadersMessage-ID: <2041426187.0.1783529294627@app132048.aus191.service-now.com>
X-ServiceNow-Source: FlowDesigner-383e6f08ce424f10ff116abb91613ce9
X-ServiceNow-Generated: true
X-ServiceNow-SysEmail-Version: 2

UpdateDelete

UpdateDelete

8.8 Flow Activation

After configuring all workflow actions, the flow is activated.

Purpose

Activation makes the workflow available for execution whenever a new Borrow Request is created.

Working

1. Validate the workflow configuration.
2. Activate the flow.
3. The workflow becomes available to all users.

Once activated, the workflow executes automatically whenever its trigger condition is satisfied.

Workflow Studio

Borrow Request Approval Flow • Smart Library Request W...

Borrow Request Approval FlowActive

TestDeactivateActivate

TRIGGER

Click or hit enter to Expand ServiceNow Core Record Trigger for Borrow Request Created or Updated where (Status is Requested)

Borrow Request Created or Updated where (Status is Requested)

ACTIONS

Select multiple

1Ask For Approval on Borrow Request

2Update Book Record

3Update Borrow Request Record

4Send Email

Add an Action, Flow Logic, or Subflow

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Data

Collapse All

Flow Variables

Trigger - Record Created or Updated

Borrow Request RecordRecord

Changed FieldsArray/Object

Borrow Request TableTable

Run Start Time UTCDate/Time

Run Start Date/TimeDate/Time

1 - Ask For Approval

2 - Update Record

3 - Update Record

4 - Send Email

emailRecord

Action StatusObject

8.9 Flow Execution

Flow execution verifies that every configured action is performed successfully.

The workflow was tested by creating multiple Borrow Requests.

The execution results confirmed that:

- The workflow started automatically.
- Approval requests were generated.
- Book Status changed from **Available** to **Issued**.
- Borrow Request Status changed to **Approved**.
- Email notifications were sent successfully.
- The workflow completed without errors.

The successful execution demonstrates that the workflow automation functions correctly and satisfies all project requirements.

Workflow Studio

Borrow Request Approval Flow

Borrow Request Approval Flow

EXECUTION DETAILS

Borrow Request Approval Flow

Test Run - Completed

Open flow

Open context record

Show Action Details

State

Start time

FLOW STATISTICS

Run as: System Administrator

Open flow logs

Completed

2026-07-08 09:43:30

171ms

TRIGGER

Borrow Request Created or Updated

Open current record

ACTIONS

1

Ask For Approval

Core Action

Completed

2026-07-08 09:43:30

111ms

2

Update Record

Core Action

Completed

2026-07-08 09:48:41

11ms

3

Update Record

Core Action

Completed

2026-07-08 09:48:41

8ms

4

Send Email

Completed

2026-07-08 09:48:41

41ms

ERROR HANDLER

CHAPTER 10

Reports and Analytics

10.1 Introduction

Reporting and analytics are essential components of any enterprise application because they transform operational data into meaningful information that supports decision-making. In ServiceNow, reports provide a visual representation of application data, allowing administrators and business users to monitor activities, identify trends, and evaluate system performance.

In the Smart Library Request Workflow application, reports were developed to analyze borrowing activity and identify the books that are most frequently borrowed by students. These reports help librarians understand user preferences, monitor library resource utilization, and make informed decisions regarding inventory management.

The project includes two graphical reports:

- Most Borrowed Books – Bar Chart
- Most Borrowed Books – Pie Chart

Both reports are generated from the **Borrow Request** table and use the **Book** field to group borrowing records.

10.2 Purpose of Reports

The reporting module was implemented to provide librarians with meaningful insights into borrowing activities. Instead of manually reviewing individual borrow request records, librarians can analyze summarized information through graphical reports.

The objectives of the reporting module are:

- To identify the most frequently borrowed books.
- To analyze borrowing patterns.
- To monitor library resource utilization.
- To assist librarians in inventory planning.
- To support data-driven decision-making.

The reports provide a quick and easy way to visualize borrowing statistics without manually examining large volumes of records.

10.3 Report Configuration

The reports were created using the ServiceNow Reporting module.

Report Details

The report counts the number of borrow requests associated with each book and displays the results graphically.

Property	Value
Report Title	Most Borrowed Books
Data Source	Borrow Request Table
Group By	Book
Aggregation	Count
Purpose	Display the number of borrow requests for each book

10.4 Most Borrowed Books – Bar Chart

The Bar Chart report provides a comparative view of the number of borrow requests for each book. Each bar represents a specific book, while the height of the bar indicates how many times that book has been borrowed.

The Bar Chart is particularly useful for comparing multiple books simultaneously and identifying which titles are most frequently requested.

Advantages

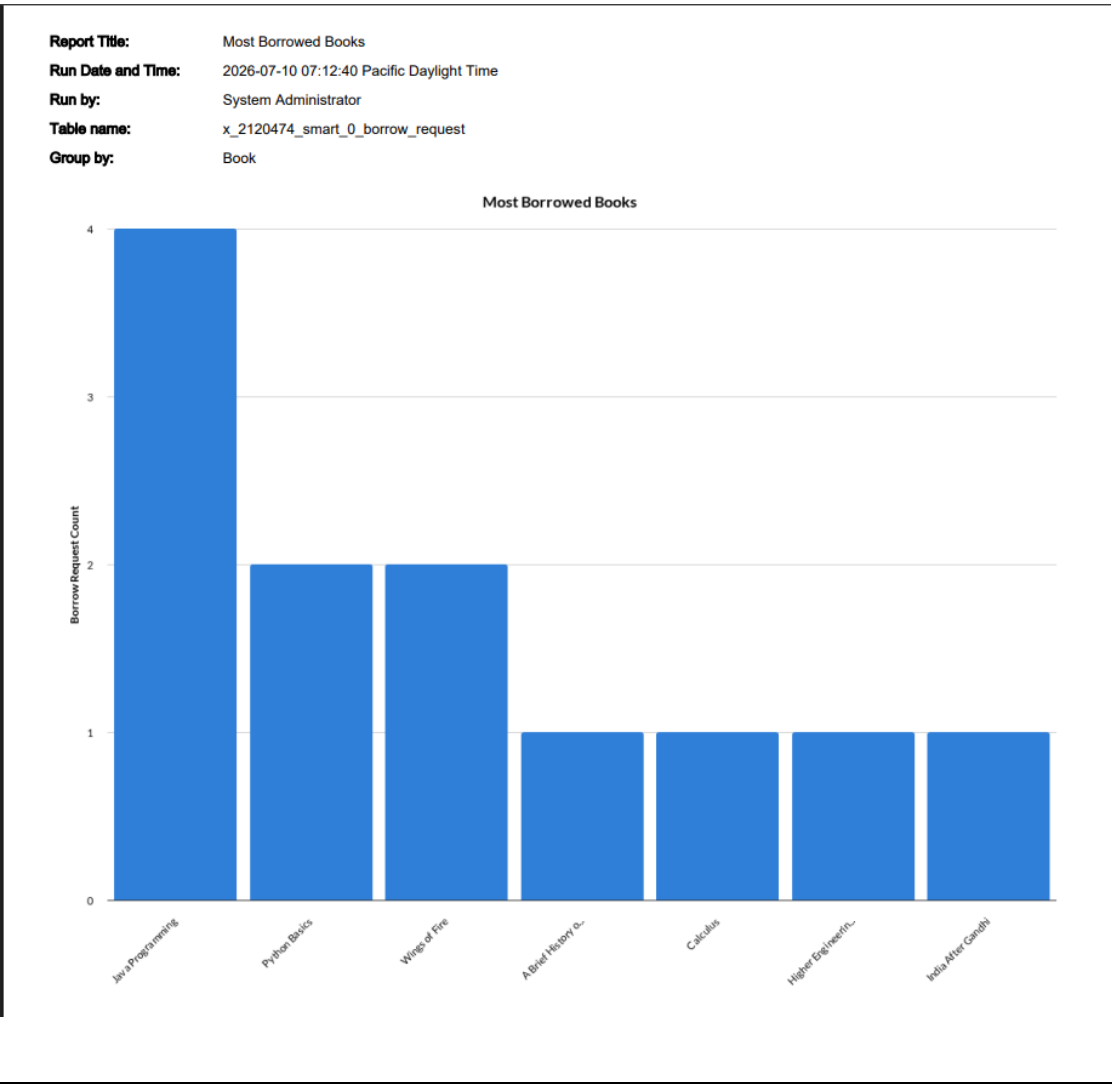
- Easy comparison between books.
- Clearly displays borrowing frequency.
- Supports inventory planning.
- Helps identify popular books.
- Improves decision-making.

Interpretation

The report enables librarians to determine:

- Which books are borrowed most frequently.
- Which books require additional copies.
- Which books are rarely borrowed.

This information helps improve library resource allocation.



10.5 Most Borrowed Books – Pie Chart

The Pie Chart provides a percentage-based visualization of borrowing activity.

Each slice of the chart represents a book, while the size of the slice corresponds to the percentage of total borrow requests associated with that book.

The Pie Chart provides an overall view of borrowing distribution and makes it easier to understand the relative popularity of different books.

Advantages

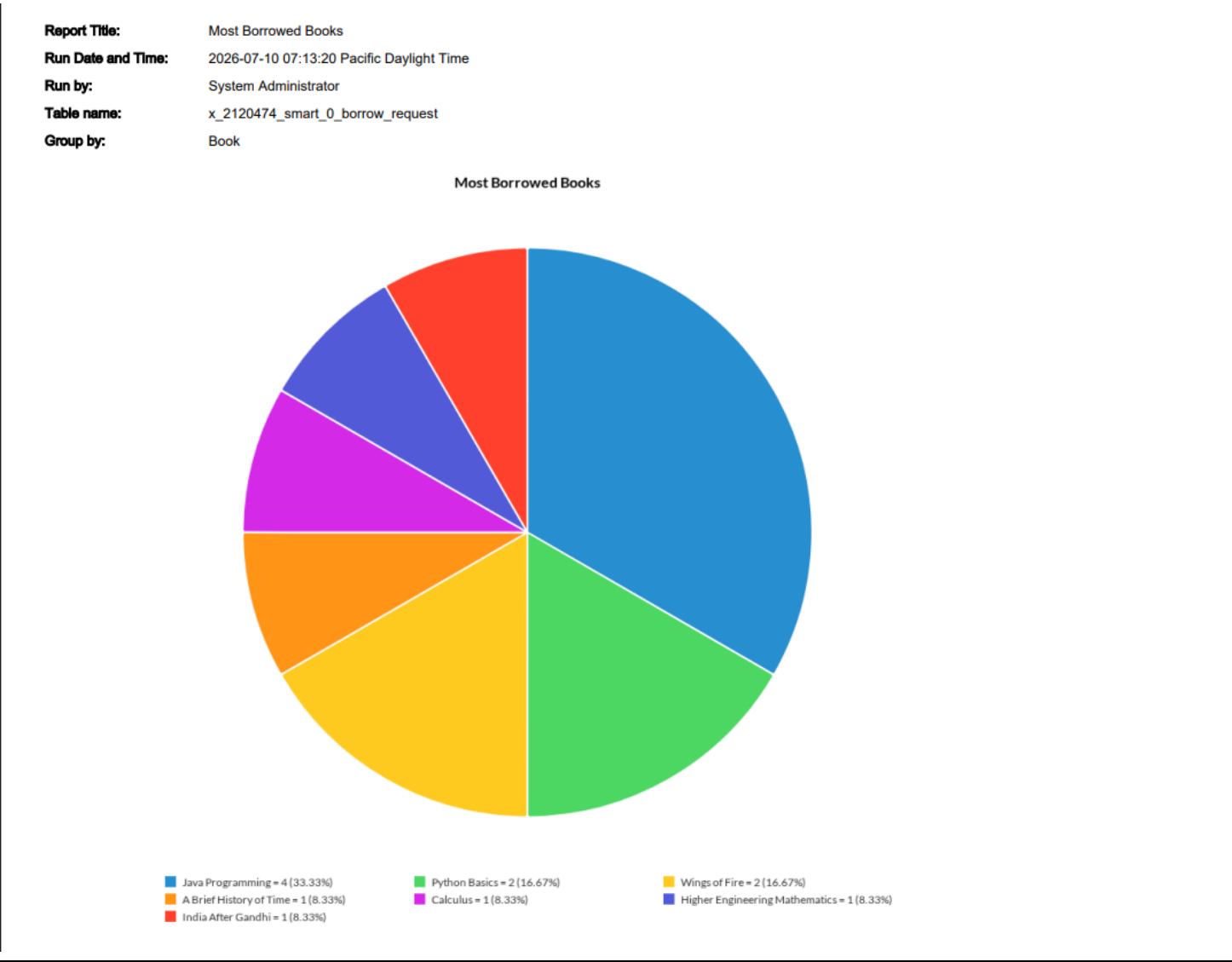
- Easy visualization of borrowing percentages.
- Attractive graphical representation.
- Quick identification of popular books.
- Suitable for presentations and reports.

Interpretation

The Pie Chart helps librarians understand:

- Overall borrowing distribution.
- Relative popularity of each book.
- Percentage contribution of each title.

The chart presents borrowing trends in an intuitive and visually appealing format.



CHAPTER 11

Testing and Validation

11.1 Introduction

Testing is an essential phase in the software development life cycle that ensures the developed application functions according to the specified requirements. The objective of testing is to verify that every module operates correctly, data is processed accurately, workflows execute successfully, and users experience the intended functionality without errors.

For the Smart Library Request Workflow application, comprehensive testing was performed after implementing all modules. Each feature, including book management, borrow request creation, workflow automation, approval process, security, reporting, and email notifications, was tested individually as well as through end-to-end workflow validation.

The testing process confirmed that the application satisfies the functional requirements and performs reliably under normal operating conditions.

11.2 Testing Objectives

The primary objectives of testing were:

- To verify that all application modules function correctly.
- To ensure the workflow executes automatically.
- To validate the approval process.
- To verify automatic updates of book status.
- To ensure email notifications are generated successfully.
- To validate security implementation using ACLs.
- To verify the Reference Qualifier filters only available books.
- To confirm the UI Policy behaves correctly.
- To validate report generation.
- To ensure data consistency throughout the application.

11.3 Testing Environment

The application was tested in the ServiceNow Personal Developer Instance (PDI).

Testing Environment

Testing was performed using different user roles to validate security and workflow execution.

Component	Description
Platform	ServiceNow Personal Developer Instance
Development Tool	ServiceNow Studio
Workflow Engine	Flow Designer
Browser	Google Chrome
Operating System	Windows 11
Database	ServiceNow Platform Database

11.4 Functional Testing

Functional testing was conducted to verify that every feature of the application performs according to the specified requirements.

The following modules were tested:

- Book Management
- Borrow Request Creation
- Approval Workflow
- Book Status Update
- Borrow Request Status Update
- Email Notification
- Reference Qualifier
- UI Policy
- Access Control Lists
- Reports

11.5 Test Cases

Test Case 1 – Book Creation

Test Case ID	TC-01
Module	Book Management
Objective	Verify book records can be created successfully
Input	Book details
Expected Result	Book record created
Actual Result	Book record created successfully
Status	☒ Pass

Test Case 2 – Borrow Request Creation

Test Case ID	TC-02
Module	Borrow Request
Objective	Verify students can create borrow requests
Input	Available Book
Expected Result	Borrow Request created
Actual Result	Borrow Request created successfully
Status	☒ Pass

Test Case 3 – Workflow Trigger

Test Case ID	TC-03
Module	Flow Designer
Objective	Verify workflow starts automatically
Expected Result	Flow executes
Actual Result	Flow executed successfully
Status	☒ Pass

Test Case 4 – Approval Process

Test Case ID	TC-04
Module	Approval
Objective	Verify librarian approval
Expected Result	Request approved
Actual Result	Request approved successfully
Status	☒ Pass

Test Case 5 – Book Status Update

Test Case ID	TC-05
Module	Book Table
Objective	Verify Book Status changes to Issued
Expected Result	Status = Issued
Actual Result	Book Status updated successfully
Status	☒ Pass

Test Case 6 – Borrow Request Status

Test Case ID	TC-06
Module	Borrow Request
Objective	Verify Borrow Request Status update
Expected Result	Status = Approved
Actual Result	Status updated successfully
Status	☒ Pass

Test Case 7 – Email Notification

Test Case ID	TC-07
Module	Notification
Objective	Verify email is sent
Expected Result	Email delivered
Actual Result	Email notification generated successfully
Status	☒ Pass

Test Case 8 – Reference Qualifier

Test Case ID	TC-08
Module	Reference Qualifier
Objective	Display only available books
Expected Result	Issued books hidden
Actual Result	Only available books displayed
Status	☒ Pass

Test Case 9 – UI Policy

Test Case ID	TC-09
Module	UI Policy
Objective	Verify UI behavior
Expected Result	Mandatory fields enforced
Actual Result	UI Policy executed correctly
Status	☒ Pass

Test Case 10 – Reports

Test Case ID	TC-10
Module	Reports
Objective	Generate borrowing statistics
Expected Result	Reports displayed
Actual Result	Bar Chart and Pie Chart generated successfully
Status	☒ Pass

CHAPTER 12

Results, Conclusion, GitHub Repository, Demo Videos and References

12.1 Results

The Smart Library Request Workflow application was successfully designed, developed, implemented, and tested using the ServiceNow platform. The project achieved all the objectives defined during the planning phase and demonstrated the effective use of ServiceNow's low-code development capabilities for automating library management processes.

The application successfully automates the complete lifecycle of a library borrow request. Students can submit borrowing requests through a user-friendly interface, while librarians can review and approve those requests using an automated approval workflow. The implemented Flow Designer automatically updates the Book and Borrow Request records, changes the availability status of books, and sends email notifications to users after approval.

Role-Based Access Control (RBAC) was implemented using ServiceNow Roles and Access Control Lists (ACLs), ensuring secure access to application resources. The Reference Qualifier successfully restricts the selection of books to only those currently available, while the UI Policy enhances data validation by enforcing mandatory fields under specific conditions.

The application also includes reporting capabilities that allow librarians to monitor borrowing trends through graphical reports such as the **Most Borrowed Books** Bar Chart and Pie Chart.

Overall, the project demonstrates how ServiceNow can be used to automate business processes, improve operational efficiency, and provide a secure and scalable application for library management.

12.2 Project Achievements

The Smart Library Request Workflow successfully achieved the following objectives:

- Developed a complete ServiceNow application using ServiceNow Studio.
- Designed custom Book and Borrow Request tables.
- Implemented workflow automation using Flow Designer.
- Created an automated approval process for borrow requests.
- Configured automatic book status updates.
- Implemented automatic Borrow Request status updates.
- Configured email notifications.
- Implemented Role-Based Access Control (RBAC).
- Applied Access Control Lists (ACLs).
- Configured UI Policies.
- Implemented Reference Qualifiers.
- Generated analytical reports.
- Successfully tested and validated the complete workflow.
- Documented the project with GitHub and demonstration videos.

These achievements demonstrate practical knowledge of enterprise application development using the ServiceNow platform.

12.3 Conclusion

The Smart Library Request Workflow successfully demonstrates how ServiceNow's low-code platform can be utilized to automate real-world business processes. By integrating workflow automation, role-based security, record management, reporting, and notifications, the application provides an efficient solution for managing library borrowing operations.

The project significantly reduces manual effort by automating repetitive tasks such as request approval, record updates, and user notifications. The use of Flow Designer ensures consistent execution of the borrowing process, while ACLs, UI Policies, and Reference Qualifiers enhance security, data integrity, and user experience.

The reporting module enables librarians to analyze borrowing trends and supports better decision-making through graphical reports. The modular architecture of the application also makes it easy to extend the system with additional functionalities in the future.

This project provided valuable practical experience in developing enterprise applications using ServiceNow and strengthened understanding of workflow automation, security implementation, and application development best practices.

12.4 Skills Acquired

During the development of this project, the following technical skills were acquired and strengthened:

ServiceNow Development

- ServiceNow Studio
- Custom Application Development
- Table Design
- Form Design

Workflow Automation

- Flow Designer
- Approval Flows
- Record Updates
- Email Notifications

Security

- User Roles
- Access Control Lists (ACLs)
- Reference Qualifiers
- UI Policies

Reporting

- ServiceNow Reports
- Bar Charts
- Pie Charts
- Data Analysis

Documentation

- Technical Documentation
- GitHub Repository Management

- Project Presentation
- Workflow Documentation

12.5 GitHub Repository

The complete project has been documented and maintained using GitHub.

The repository contains:

- Project Documentation
- Architecture Diagram
- Screenshots
- Reports
- README File
- Demo Video Link

The GitHub repository enables version control, project sharing, and future enhancements.

GitHub Repository

<https://github.com/NandaKishore-04/Smart-Library-Request-Workflow-ServiceNow>

12.6 Project Demonstration

The project demonstration videos have been uploaded to Google Drive for easy access.

The demonstration includes:

- Smart Library Request Workflow Demo
- Borrow Request Creation
- Borrow Request Approval Workflow

Demo Video Link

https://drive.google.com/drive/folders/1DKtuXi0ce_UnRyG2awF13NEzkjMuGnyF?usp=drive_link

12.7 References

The following resources were used during the development of this project:

1. ServiceNow Documentation
<https://docs.servicenow.com>
2. ServiceNow Developer Portal
<https://developer.servicenow.com>
3. ServiceNow Community
<https://www.servicenow.com/community>
4. Flow Designer Documentation
<https://docs.servicenow.com>

5. ServiceNow Studio Documentation
<https://developer.servicenow.com>
6. GitHub Documentation
<https://docs.github.com>
7. Google Drive Documentation
<https://support.google.com/drive>

12.8 Overall Project Summary

The Smart Library Request Workflow successfully automates library operations using ServiceNow's enterprise platform. The project integrates custom application development, workflow automation, role-based security, reporting, and notifications into a single scalable solution.

Throughout the internship, practical experience was gained in designing database tables, configuring workflows, implementing security mechanisms, creating reports, and documenting enterprise applications. The completed project demonstrates the effective use of ServiceNow for solving real-world business problems through low-code development.